

A Blockchain-Driven Hierarchical Authentication and Key Agreement Scheme for VANETs With Cloud-Edge Collaboration

Lu Wei¹, Member, IEEE, Yongjuan Zhang, Jie Cui², Senior Member, IEEE, Hong Zhong³, Irina Bolodurina⁴, and Debiao He⁵, Member, IEEE

Abstract—Vehicular ad-hoc networks (VANETs) are the cornerstone of intelligent transportation systems, designed to enhance road safety and traffic efficiency. However, their dynamic and distributed nature poses significant challenges for secure communication and key management. Traditional authentication and key agreement (AKA) schemes for VANETs often rely on centralized trust architectures, resulting in system security and reliability issues. Despite the introduction of distributed trust architecture schemes that have appeared recently, they fail to solve one issue, i.e., how the key agreement requests can be authenticated in the distributed communication scenario where the authentication authorities are all non-full-credible and have differentiated credibility. To solve this issue, we propose a hierarchical AKA scheme for VANETs with cloud-edge collaboration powered by consortium blockchain. Specifically, we first proposed a vehicle reputation evaluation algorithm for evaluating the trustworthiness of the vehicle, so that the AKA requests sent by vehicles with low reputation will be rejected. On the basis of the reputation evaluation algorithm, we proposed a hierarchical threshold-based AKA scheme for VANETs where cloud servers (CSs) and edge servers (ESs) can collaboratively authenticate the AKA requests, so that the authentication service can be trusted upon getting authenticated by a series of valid combinations of CSs and ESs. Both formal and informal security proofs validate the security of our proposed scheme, and simulation experiments demonstrate its efficiency.

Index Terms—Authentication and key agreement, hierarchical authentication, vehicular ad-hoc networks, blockchain.

Received 24 September 2024; revised 1 April 2025; accepted 12 June 2025. Date of publication 18 June 2025; date of current version 27 June 2025. This work was supported in part by the National Natural Science Foundation of China under Grant U23A20308, Grant 62302008, Grant 62202008, and Grant 62325209; in part by the Natural Science Foundation of Anhui Province, China, under Grant 2408085JX010; and in part by the University Synergy Innovation Program of Anhui Province under Grant GXXT-2022-049. The associate editor coordinating the review of this article and approving it for publication was Prof. Kun Sun. (Corresponding author: Jie Cui.)

Lu Wei, Yongjuan Zhang, Jie Cui, and Hong Zhong are with the Key Laboratory of Intelligent Computing and Signal Processing of Ministry of Education, School of Computer Science and Technology, and the Institute of Physical Science and Information Technology, Anhui University, Hefei 230601, China, and also with Anhui Engineering Laboratory of IoT Security Technologies, Anhui University, Hefei 230039, China (e-mail: cuijie@mail.ustc.edu.cn).

Irina Bolodurina is with the Faculty of Mathematics and Information Technologies, Orenburg State University, 460018 Orenburg, Russia (e-mail: prmat@mail.osu.ru).

Debiao He is with the School of Cyber Science and Engineering, Wuhan University, Wuhan 430072, China, and also with Shanghai Key Laboratory of Privacy Preserving Computation, MatrixElements Technologies, Shanghai 201204, China (e-mail: hedeiao@163.com).

Digital Object Identifier 10.1109/TIFS.2025.3581022

I. INTRODUCTION

THE vehicular ad-hoc networks (VANETs) are an open converged network that is dedicated to enhancing automotive intelligence and autonomous driving through the all-round interconnection of vehicles, people, and networks, as well as innovating the automotive and transportation service industry to enhance transportation efficiency and elevate the overall driving experience.

With the rapid increase in vehicles and diverse user demands, vehicle-generated data is soaring. Effective data processing is crucial for a secure and efficient vehicular network, resulting in traditional VANETs communication architectures always rely on centralized cloud servers (CSs) to process data. However, while cloud servers (CSs) handle most data computation and storage, their distance from vehicles can cause high latency. Additionally, centralized VANETs communication architectures are always vulnerable to various security attacks, especially for internal attacks. For instance, Tesla suffered a severe data breach orchestrated by an internal adversary, where a whistleblower allegedly accessed and leaked approximately 100GB of sensitive and proprietary information such as customer data, employee salaries, bank accounts, and source code [1]. Likewise, Toyota encountered a production shutdown in Japan when a trusted internal supplier became the entry point for a cyberattack, prompting the suspension of operations in 14 factories for a full day [2].

In order to address the performance and security issues inherent in centralized VANETs communication architectures, the decentralized and cloud-edge collaborative architecture has been proposed recently. By leveraging the powerful computation capabilities of cloud servers alongside the proximity advantages of edge servers (ESs), which enable real-time data processing closer to the data source, this approach ensures both security and high performance for VANETs communications. Furthermore, the cloud-edge collaborative vehicular networking framework improves service quality for real-time traffic applications, resolving the shortcomings of purely cloud-based systems and guaranteeing timely, reliable services for vehicles and users.

Under distributed VANETs communication architectures, distributed authentication and key agreement (AKA) schemes designed for the secure communication requirements of

VANETs have shown rapid growth. Zhang et al. [3] introduced an innovative distributed aggregate AKA scheme. The scheme adopts a layered trusted architecture. While this scheme effectively ensures system robustness, it requires RSUs to be honest and trustworthy, leading to high computational and communication overhead. Zhu and Liu [4] designed an AKA scheme using chaotic mapping that supports a multi-server-to-server architecture in which any server can assume the role of a registry. Thus, this approach necessitates attention to avert potential single points of failure.

While existing AKA schemes can ensure security in distributed key agreement processes under a single authenticated entity and effectively mitigate external attacks, they exhibit critical limitations in complex distributed communication scenarios where authenticated entities possess heterogeneous privileges and may not be fully trustworthy. To overcome these challenges, we propose a novel blockchain-driven hierarchical AKA scheme for vehicular ad hoc networks (VANETs) with cloud-edge collaboration. The proposed scheme innovatively leverages cloud servers and edge servers as hierarchically structured authentication entities with differentiated permission levels to collaboratively process AKA requests. Within this framework, an AKA request is deemed authenticated only when the combined authentication shares from both the cloud and edge servers satisfy a predefined threshold, thereby substantially bolstering the security of the authentication process. To further fortify the system's resilience against internal attacks, the scheme incorporates a dynamic reputation evaluation mechanism, which synthesizes a vehicle's historical communication behavior with real-time communication data to dynamically assess and update the vehicle's reputation score. If a vehicle's reputation falls below a system-defined security threshold, it is promptly excluded from participating in VANETs communications, thereby preemptively neutralizing potential insider threats. The main contributions are summarized as follows:

- 1) We design a threshold-based hierarchical authentication AKA scheme for securing VANETs. The scheme uses CSs and ESs to build an authentication network for complex distributed scenarios in which authentication entities have different privileges and are not fully trusted, which not only eliminates the dependence on trusted centres, but also gets rid of the dependence on the master key through the concept of hierarchical threshold voting. Even if a single server fails, the security of the system will not be affected, thus significantly improving the security and robustness of the overall system.
- 2) We propose a dynamic vehicle reputation mechanism to enhance the security of AKA sessions. The mechanism dynamically updates the reputation value of a vehicle by evaluating its communication behavior in real-time and imposes reputation penalties on vehicles sending false or invalid information. When a vehicle's reputation value falls below a threshold set by the system, the vehicle will be prohibited from launching AKA requests, thus greatly reducing the probability of the engagement of internal adversaries in the AKA session.

- 3) Finally, comprehensive security proofs and thorough analyses are conducted. Experimental outcomes validate that our scheme effectively fulfills the security objectives of vehicular networks, offering distinct security benefits over the latest VANETs AKA schemes.

II. RELATED WORK

The AKA scheme plays a crucial role in protecting vehicular communication and addresses two core issues. First, it ensures the identity of the participants and the authentication of the interaction messages used for key agreement. Second, it ensures the correctness and security of the key agreement process. Since the communication channel of VANET is open and insecure, it becomes crucial to establish an AKA mechanism to support the legitimacy of the participants.

Currently, most of the AKA schemes in VANET rely on a centralized trusted architecture, which necessitates a fully trusted authority center to undertake functions such as registration, authentication, and revocation for participants including vehicles and roadside units. Vijayakumar et al. [5] devised a key agreement scheme based on batch authentication, which effectively reduces the computational cost at the vehicle side by constructing a batch authentication algorithm executed by the RSU. Ma et al. [6] devised an efficient ECC-based AKA scheme with trusted center involvement for mutual authentication between vehicles and fog units, ensuring secure communication within the system. Abbasinezhad-Mood et al. [7] introduced an innovative key agreement protocol tailored for the vehicle-to-grid network, emphasizing anonymity and efficiency. Their scheme leverages chaotic mapping to facilitate secure and anonymous session key establishment between electric vehicles and aggregators, enhancing the protection of this network.

Wei et al. [8] presented a tree-structured AKA protocol designed to safeguard both V2V and V2I communications simultaneously. This protocol optimizes the process of vehicle joining or leaving the network, minimizing both computational and communication costs, thereby improving overall efficiency and scalability. Chuang and Chen [9] introduced a multi-server AKA scheme that offers anonymity to users, enabling a single registration process for access to multiple servers. However, the dependency on a centralized registry poses a potential risk of single-point-of-failure. Cui et al. [10] introduced an adaptable and resilient key exchange mechanism tailored for multi-cloud setups. Nevertheless, its reliance on a solitary TA, renders it susceptible to DoS attacks. Islam et al. [11] devised a lightweight approach for group key agreement, emphasizing minimal computational burden. However, it suffers from reduced practicality due to numerous one-to-one interactions and has security flaws like vulnerability to man-in-the-middle attacks and lack of forward security.

To reduce reliance on trusted centers, schemes based on distributed trust architectures, particularly leveraging blockchain technology have emerged. These schemes aim to achieve authentication [12], [13] and trust management [14], [15] for IoT communication, enhancing system security and robustness.

Ma et al. [16] introduced a decentralized key agreement mechanism leveraging blockchain technology for efficient registration, updates, and revocation of vehicle public keys, ensuring authenticity validation. However, only the vehicle service provider and the RSU can trigger smart contracts on the blockchain network, limiting decentralization to a local level. Zhang et al. [17] devised a blockchain-integrated AKA scheme, facilitating the secure exchange of key agreement messages through blockchain transactions. However, the chained loop structure causes delays as vehicles must wait for block confirmations sequentially, and the use of bilinear pair cryptography adds further delays unsuitable for real V2X communication scenarios. Meng et al. [18] proposed a private chain-based AKA scheme for secure vehicle-edge server communication. It records shared keys in the private chain and introduces a public key registry for key management. However, the scheme's private chain network nodes are vulnerable to attacks, compromising its security. Li et al. [19] introduced a blockchain-based certifiable key agreement protocol, emphasizing unlinkability. However, the scheme relies on the consortium blockchain, cannot run on the public chain, and involves frequent homomorphic cryptographic operations, leading to large computational and communication overheads.

Xu et al. [20] proposed an efficient AKA scheme using blockchain. Trusted centers maintain the network and authentication algorithms. Blockchain stores vehicle identity for cross-center authentication. However, it assumes trusted centers are honest and require security tamper-resistant devices, posing high-security assumptions. Furthermore, reputation systems have garnered significant attention in recent vehicular network advancements, with various researchers proposing systems to uphold message trustworthiness. Li et al. [21] introduced a mechanism enabling vehicles to gauge message reliability by assessing senders' reputation scores. These scores reflect the vehicle's historical track record of credible messaging, thereby providing insights into the likelihood of future credible contributions. Yang et al. [22] proposed a reputation system integrating blockchain technology for assessing data trustworthiness, enabling vehicles to compute senders' reputation scores based on blockchain-stored ratings. However, both this and the previously mentioned scheme fall short of addressing comprehensive vehicular network security challenges.

In summary, the research on AKA for VANET is mainly divided into two aspects: centralized and distributed trusted architectures, with the former having excellent performance but high security assumptions, and the latter facing challenges such as large computational and communication overheads and insufficient security strength although it solves some of the problems, and both of them have not fully explored the highly efficient authentication key agreement mechanism using blockchain technology under the non-trusted and semi-trusted environments.

III. PRELIMINARIES AND BACKGROUND

A. Elliptic Curve Cryptosystem

In the field of VANET security research, the elliptic curve cryptosystem (ECC) is a commonly used encryption technique

that leverages the mathematical properties of elliptic curves to provide security and efficiency. We introduce ECC from the following perspectives:

1) *Selection of Finite Field and Elliptic Curve*: We first define a finite field F_p , where p is a prime number that specifies the field's size. Subsequently, we select an elliptic curve E defined over this finite field F_p , typically represented by $y^2 = x^3 + ax^2 + b \mod p$, where a and b are elements in the finite field F_p .

2) *Group Structure and Generator*: In the ECC, we introduce a cyclic additive group G , where P serves as the generator of the group. The prime order q of the group G determines the complexity and security of the ECC.

3) *Computational and Decisional Challenges*:

- **Elliptic Curve Discrete Logarithm Problem (ECDLP)**: On an elliptic curve E , if given two points P and Q such that $Q = sP$, the likelihood of computing s for a probabilistic polynomial-time (PPT) adversary is considered negligible.
- **Elliptic Curve Decisional Diffie-Hellman Problem (ECDHP)**: On an elliptic curve E , if given two tuples P, xP, yP, xyP and P, xP, yP, zP , the likelihood of distinguishing between these two tuples for a PPT adversary is considered negligible.

B. Secret Sharing Schemes

A (t, n) secret sharing scheme [23] over Z_q is a polynomial interpolation based threshold secret sharing scheme proposed by Adi Shamir in 1979. The core idea is to partition the secret S into n shares and set a threshold value k ($1 < k \leq n$) such that the secret S cannot be recovered by satisfying any less than k shares and can be uniquely determined by any less than k shares. The core principle is as follows:

First, construct a $(k-1)$ -order polynomial $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_{k-1}x^{k-1}$ where $a_0 = S$ (secret) and the factors $a_1 \dots a_{k-1}$ are random numbers. Then, shares $(x_i, f(x_i))$ ($i = 1, 2, \dots, n$) are generated for n participants, with x_i being the public non-zero unique identifier and $f(x_i)$ being the private share value. The secret $S = a_0$ can only be obtained by reconstructing the polynomial to compute $f(0)$ using Lagrange interpolation as shown in Equation (1) if at least k secret shares are collected.

$$f(x) = \sum_{i=1}^k f(x_i) \prod_{j=1, j \neq i}^k \frac{x - x_j}{x_i - x_j} \quad (1)$$

C. System Model

Our proposed scheme involves five types of entities which are the system initiator, cloud server (CS), edge server (ES), roadside unit (RSU), and vehicle, that collaborate within the system to achieve its overall functionality. The following is a detailed description of each type of entity and the system model visually represented in Fig. 1.

- **System initiator**: The system initiator is mainly responsible for the generation of system and security parameters, the selection of semi-trusted entities that have the privilege of authenticating vehicles, and the management

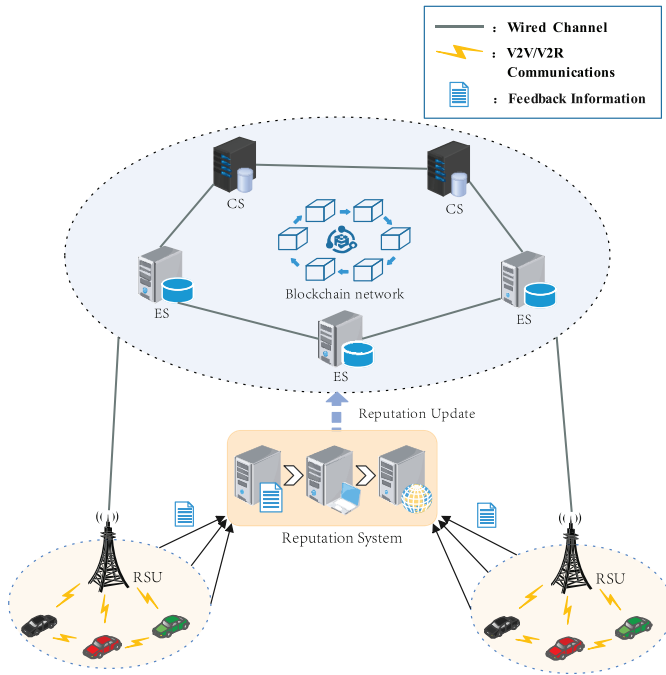


Fig. 1. System model.

of the blockchain network. Note that once the system initiator is not involved in the AKA process.

- **CS:** CSs are semi-trusted entities deployed in the cloud, leveraging their computational and storage capabilities primarily to contribute to the construction of the consortium blockchain, and achieving the authentication of the AKA request message in the authentication phase. And are also responsible for offering registration services for vehicles and RSUs.
- **ES:** ESs are semi-trusted entities deployed in closer physical distance, which are mainly involved in the authentication of AKA request messages in this scheme.
- **RSU:** The RSU primarily facilitates network access for vehicles and disseminates traffic and entertainment updates to neighboring vehicles. In our proposed scheme, the RSU collaborates with vehicles to initiate a key agreement request.
- **Vehicle:** Vehicles are the most common entities in the vehicular network. We set reputation values for vehicles to assess their historical behaviour, as they are vulnerable to internal or external attacks. The legitimacy of the vehicle and the messages sent by the vehicle will be authenticated based on the reputation value of the vehicle.

IV. THE PROPOSED SCHEME

This section presents a comprehensive overview of our proposed framework, structured into five distinct phases: setup, reputation mechanism, registration, request, and response. The steps involved in each phase are detailed below and the main procedures are as shown in Fig. 2.

A. Setup Phase

In this phase, the consortium blockchain is constructed, and system parameters are generated, along with the selection of participating entities.

1) *Blockchain Construction:* A consortium blockchain, jointly administered and operated by multiple entities, boasts of enhanced efficiency, robust security measures, and a decentralized structure. In our proposed scheme, CS and ES jointly construct and maintain a distributed consortium blockchain network for authentication. We utilize the PBFT consensus mechanism to safeguard the system's reliability and security. CS and ES participate in the authentication process of requesting entities independently, respectively. A more detailed explanation of this configuration will be presented in subsequent sections of our work.

2) *Parameter Initialization:* The initialization phase mainly involves the selection of basic parameters and functions. CS and ES cooperate to build the consortium blockchain and complete the initialization process without relying on trusted entities. Due to the differences in computing and storage capacity between CS and ES, they will be given different secret sharing shares to participate in authentication during the hierarchical threshold secret sharing phase. The specific steps of the initialization process are outlined below:

- 1) The system initiator chooses a secure elliptic curve $E : y^2 = x^3 + ax + b \mod p$. A point P can be chosen from E to generate an additive elliptic curve group G with order q . In addition, system thresholds t_1, t_2 are chosen to balance security and performance. In the key agreement phase, e.g., it is necessary to satisfy that the total authentication messages are no less than t_2 and no less than t_1 of them come from the CS.
- 2) It is assumed that the number of CSs and ESs in the consortium blockchain are n_1, n_2 , where $n_1 + n_2 = n$. j and l are the serial numbers of CSs and ESs respectively ($1 \leq j \leq n_1, 1 \leq l \leq n_2$). First, each CS and ES entity selects random numbers $x_j, x_l \in \mathbb{Z}_q$ to calculate their respective addresses $addr_j, addr_l$ and sends the registration request ciphertext that containing the address information to the registration authority (RA). Upon receiving the registration request, RA triggers the execution of the smart contract function detailed in Algorithm 1 via a signed transaction. Here $tag1$ is used as a tag value to distinguish the type of server, with $tag1 = 0$ denoting CS, and $tag1 = 1$ denoting ES, respectively. Each CS and ES entity obtains a valid blockchain address upon completing the registration.
- 3) Next, each CS generates random numbers $a_j, b_j \in \mathbb{Z}_q$ and each ES generates $b_l \in \mathbb{Z}_q$ as their respective secret values. They then compute the public keys $U_j = a_j P$, $K_j = b_j P$, $K_l = b_l P$. The blockchain network secures the storage of each entity's public key information through the smart contract functions outlined in Algorithm 3. Here we leverage a cuckoo filter to efficiently store the fingerprints of public keys, thereby minimizing the associated storage overhead. Next, each server generates polynomials in the form of Equations (2), where

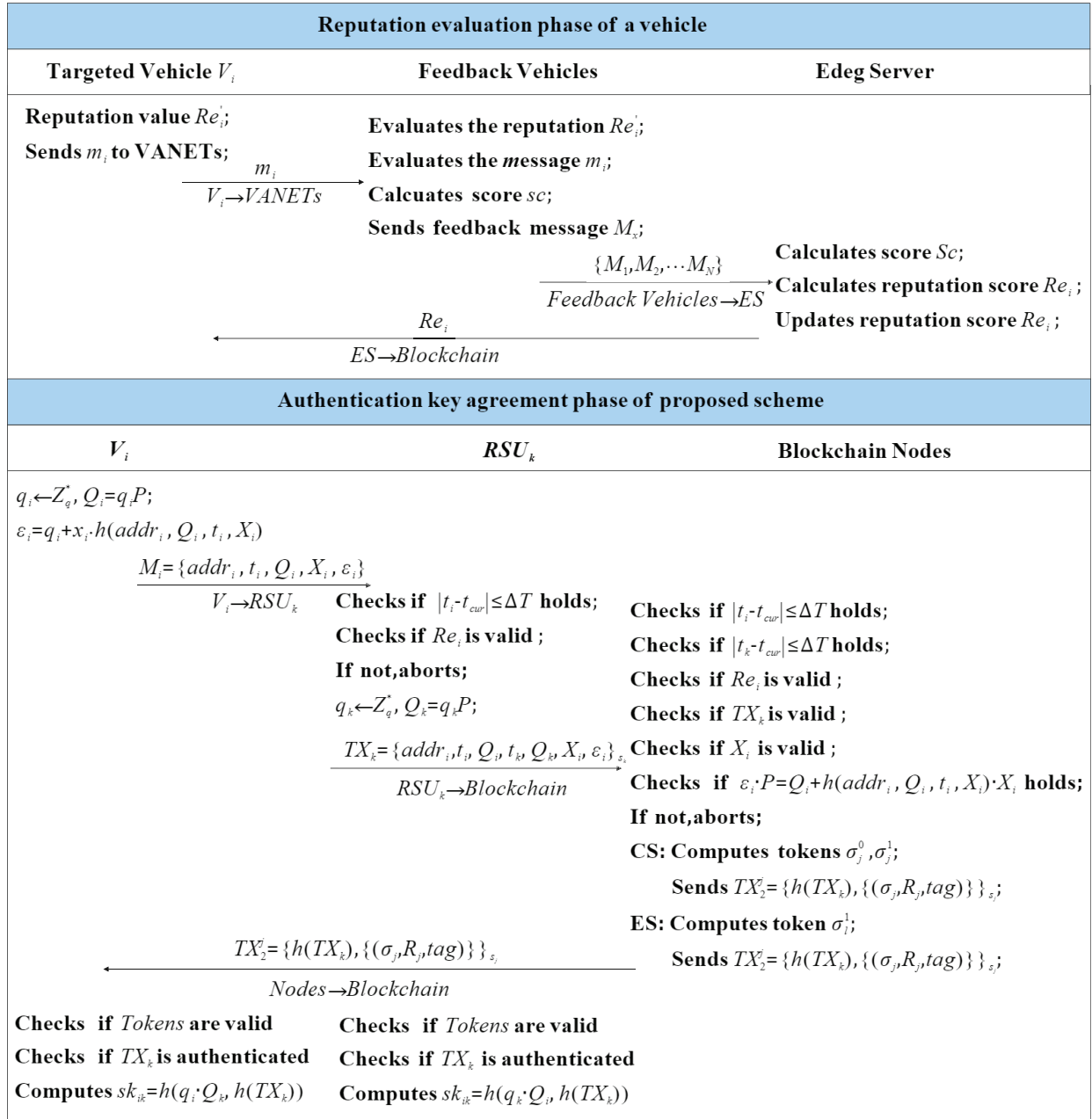


Fig. 2. The main procedures of our proposed scheme.

$a_{j,1} \dots a_{j,t-1}$ are random coefficients, u_j is the server's corresponding secret value, and t is the system's pre-defined threshold. Specifically, each CS generates two polynomials, $h_j(x)$ and $p_j(x)$, from two secret values, with thresholds t taken as t_1 and t_2 respectively (where $t_1 < t_2$). Meanwhile, the ES generates a polynomial $p_l(x)$ with a threshold of t taken as t_2 .

$$f_j(x) = u_j + a_{j,1}x + \dots + a_{j,t-1}x^{t-1} \mod q \quad (2)$$

- 4) Each CS and ES obtains the set of secret shares ss_j and ss_l of their respective polynomials, as shown in Equation (3) and Equation (4). In the secret sharing phase, each server shares the encrypted and signed secret share $ss_{j,k} = (k, f_j(k))$ with other servers, where k is

the sequence number of the corresponding server and $k \neq j$. After completing the above operations, each CS computes two collaborative private keys $sk_{0,j}$ and $sk_{1,j}$ as shown in Equation (5) and Equation (6) from the secret shares of polynomials $h(x)$ and $p(x)$, respectively, and each ES computes one collaborative private key $sk_{1,l}$ as shown in Equation (7) from the secret share of polynomial $p(x)$.

$$ss_j = \{(idx_0, h_j(idx_0)) \cup (idx_1, p_j(idx_1)) \mid idx_0 \in [1, n_1] \ \& \ idx_1 \in [1, n] \ \& \ idx \neq j\} \quad (3)$$

$$ss_l = \{(idx, p_l(idx)) \mid idx \in [1, n] \ \& \ idx \neq l\} \quad (4)$$

$$sk_j = sk_{0,j} = h_1(j) + h_2(j) + \dots + h_{n_1}(j) \mod q \quad (5)$$

$$sp_j = sk_{1,j} = p_1(j) + p_2(j) + \dots + p_n(j) \mod q \quad (6)$$

$$sp_l = sk_{1,l} = p_1(l) + p_2(l) + \dots + p_n(l) \bmod q \quad (7)$$

Algorithm 1 ES/CS Registration Function

Input: $addr_i, tag1$
Output: *bool*

```

1: ▷ check the validity of the function caller
2: ▷ ras is an address array
3: if msg.sender not inras then
4:   return false
5: end if
6: ▷ ess and css are mapping types
7: if tag1 == 1 then
8:   ess[msg.sender][addr] ← true
9: else
10:  css[msg.sender][addr] ← true
11: end if

```

Algorithm 2 Vehicle/RSU Registration Function

Input: $addr_i$
Output: *bool*

```

1: ▷ check the validity of the function caller
2: ▷ csa is an address array
3: if msg.sender not incsa then
4:   return false
5: else
6: ▷ va is a mapping type
7: va[msg.sender][addr_i] ← true
8: return true
9: end if

```

Algorithm 3 Public Key Registration Function

Input: $addr_j, pks_i$
Output: *bool*

```

1: if vf == NULL then
2:   ▷ init an empty Cuckoo filter with specific table size
   and maximum kick counts for a type of registrant
3:   vf ← new(Size, MaxKicks)
4: end if
5: if va[addr_j][msg.sender] == True then
6:   for each  $pk_i \in pks_i$  do
7:     ▷ call the insert function of the Cuckoo filter
8:     vf.insert(pk_i)
9:   end for
10:  return true
11: else
12:   return false
13: end if

```

- 5) All public information $\{E, G, q, P, n, U_j, K_j, K_l\}$ will be stored on the consortium blockchain. Meanwhile, the system public key $PK = \sum_{j=1}^n K_j + \sum_{j=1}^{n1} U_j = K + U$ can be derived from the information on the blockchain.

B. Reputation Mechanism

Reputation value serves as an indicator for community members to evaluate each other, enabling users to gauge the

credibility of messages based on the sender's reputation value. In our framework, the ES is responsible for performing the collection, computation, and updating of reputation values. Specifically, we categorize the reputation model into three modules, i.e., the feedback collection, the reputation calculation, and the reputation update modules. The reputation value is defined as a value between an interval (e.g. [60, 100]), and user identities with reputation values below the minimum of the interval will be blacklisted and their identities will be broadcast.

1) *Feedback Collection Module*: For the message from the target vehicle, the receiver first evaluates whether the target vehicle is trustworthy based on the sender's initial reputation value (messages from vehicles blacklisted and broadcasting their identities are not accepted). Subsequently, the feedback vehicle evaluates the received message m based on the current traffic environment to provide a credibility assessment Sc for the message ($0 \leq Sc \leq 1$). Then, the feedback vehicle sends feedback information M to the feedback collection server, including the vehicle's own information, the message m , the corresponding assessment score sc , and a timestamp. All N vehicles participating in the feedback assessment constitute a feedback set.

2) *Reputation Calculation Module*: Firstly, the feedback information are divided into three reputation interval groups based on the reputation values of the feedback vehicles in the feedback set. For example, [60-75], [75-90], [90-100], where the number of feedback vehicles in each reputation interval group is n_1, n_2, n_3 ($n_1 + n_2 + n_3 = N$). The system assigns different evaluation weights w_1, w_2, w_3 for each group (where $w_1 < w_2 < w_3, w_1 + w_2 + w_3 = 1$), meaning feedback vehicles in groups with higher reputation values have higher evaluation weights. Then, the weighted credibility assessment Sc of the target vehicle is calculated using Equation (8), where s_{1k}, s_{2k} , and s_{3k} (all $\in [0, 1]$) represent the ratings given by feedback vehicles from the different reputation groups.

$$Sc = \frac{\omega_1}{n_1} \cdot \sum_{k=1}^{n_1} s_{1k} + \frac{\omega_2}{n_2} \cdot \sum_{k=1}^{n_2} s_{2k} + \frac{\omega_3}{n_3} \cdot \sum_{k=1}^{n_3} s_{3k} \quad (8)$$

Finally, based on the magnitude of the weighted credibility assessment Sc , we calculate the reputation of the vehicle. If $Sc > \beta$, we deem the message from the target vehicle to be validated by a majority of vehicles, and accordingly, we reward its reputation value using Equation (9). Conversely, if $Sc < \beta$, we deem the message from the target vehicle to be questioned by the majority of vehicles, and thus, we penalize its reputation value using Equation (10). Here β is the system's preset reputation threshold, the value of which can be set and adjusted according to the security requirements based on the VANETs system.

$$Re_i = Re'_i + r_b \cdot \left(1 - \frac{Re'_i}{Re_{max}}\right) (\beta_1 \cdot D^{\beta_3} + \beta_2 \cdot Sc^{\beta_4}) \quad (9)$$

$$Re_i = Re'_i - r_b \cdot \left(\frac{Re'_i}{Re_{min}} - 1\right) (\beta_1 \cdot D^{\beta_3} + \beta_2 \cdot (1 - Sc)^{\beta_4}) \quad (10)$$

The formula introduces two terms, $(1 - Re'_i/Re_{max})$ and $(Re'_i/Re_{min} - 1)$, which effectively constrain the rapid growth of reputation values for entities with high current reputation

values and impose more severe penalties when they publish false messages. In this context, Re'_i represents the historical reputation value of the vehicle, Re_{max} and Re_{min} represent the maximum and minimum of the credible reputation value interval, respectively, D denotes the current vehicle density and r_b stands for the base value for reputation rewards and penalties, and $\beta_1, \beta_2, \beta_3, \beta_4$ are four predefined weighting factors.

Remark: It is worth noting that the reputation mechanism is designed without bias under the condition that feedback scores s_{ik} are independent and identically distributed (i.i.d.) within each reputation group and the density factor D is properly normalized. Specifically, weight assignment unbiasedness, interval grouping unbiasedness, and stability of reward and penalty functions can be achieved, which can be proved as follows.

- 1) **Weight Assignment Unbiasedness:** Let Sc be the weighted credibility score and assume s_{ik} are i.i.d. with $E[s_{ik}] = \mu$ for any i and k , then we get Equation (11). This shows that the expected score equals the underlying true mean μ , proving the weight assignment introduces no bias.

$$\begin{aligned}
 E[Sc] &= E \left[\frac{w_1}{n_1} \sum_{k=1}^{n_1} s_{1k} + \frac{w_2}{n_2} \sum_{k=1}^{n_2} s_{2k} + \frac{w_3}{n_3} \sum_{k=1}^{n_3} s_{3k} \right] \\
 &= \frac{w_1}{n_1} \sum_{k=1}^{n_1} E[s_{1k}] + \frac{w_2}{n_2} \sum_{k=1}^{n_2} E[s_{2k}] + \frac{w_3}{n_3} \sum_{k=1}^{n_3} E[s_{3k}] \\
 &= \frac{w_1}{n_1} n_1 \mu + \frac{w_2}{n_2} n_2 \mu + \frac{w_3}{n_3} n_3 \mu = \mu
 \end{aligned} \quad (11)$$

- 2) **Interval Grouping Unbiasedness:** Consider all possible distributions of vehicles across groups G_1 (group with low reputations), G_2 (group with middle reputations), and G_3 (group with high reputations). If all vehicles belong to G_1 , G_2 , or G_3 , then $Sc \approx w_1\mu$, $Sc \approx w_2\mu$, and $Sc \approx w_3\mu$ hold in these three cases, respectively, according to Equation (11). If the distributions of vehicles across all groups, then $Sc \approx w_1\mu + w_2\mu + w_3\mu = \mu$. Therefore, the grouping mechanism maintains unbiasedness as long as $w_1 < w_2 < w_3$ (our intentional design).
- 3) **Stability of Reward and Penalty functions:** According to Equations (9, 10), the reward function can be expressed with Equation (12). By Brouwer's fixed-point theorem, since f_{reward} is continuous on the compact convex set $[Re_{min}, Re_{max}]$ and f_{reward} maps $[Re_{min}, Re_{max}]$ to itself, there exists at least one fixed point Re^* where $f(Re^*) = Re^*$. Additionally, the derivative of the reward function satisfies $\frac{\partial f_{reward}}{\partial Re_i} = 1 - \frac{r_b g(D, Sc)}{Re_{max}} \in (0, 1)$, since $g(D, Sc) = \beta_1 D^{\beta_3} + \beta_2 S c^{\beta_4} \leq 1$ where $\beta_1 + \beta_2 \leq 1$, this contraction property guarantees convergence to the fixed point (same for the penalty function).

$$f_{reward}(Re_i) = Re_i + r_b \left(1 - \frac{Re_i}{Re_{max}} \right) g(D, Sc) \quad (12)$$

- 3) **Reputation Update Module:** The ES uploads an updated reputation score to the blockchain. If the target vehicle's

reputation falls beneath a predetermined threshold, the ES blacklists it and broadcasts the vehicle's real identity.

C. Registration Phase

In the registration phase, the RSU and vehicle need to complete address and public key registration respectively. The detailed process proceeds as outlined below:

1) **RSU Registration:** Firstly, RSU_k selects random numbers $e_k, x_k \in \mathbb{Z}_q$ and computes the blockchain address $addr_k$ and public key $X_k = x_k P$. RSU_k forwards a registration request containing the real identity to the CS_j . In turn, CS_j initiates the smart contract, defined in Algorithm 2, by submitting a signed transaction $TX_j^0 = \{addr_k\}_{a_j}$ to the blockchain network. Upon confirmation of TX_j^0 by the blockchain network, the legitimacy of the RSU_k can be verified. Next, RSU_k sends the signed transaction $TX_k^1 = \{X_k\}_{e_k}$ to the blockchain network, invoking the smart contract specified in Algorithm 3 for public key registration. In this way, RSU_k completes its registration on the blockchain.

2) **Vehicle Registration:** Firstly, V_i selects the random numbers $e_i, x_i \in \mathbb{Z}_q$ and computes the blockchain address $addr_i$ and public key $X_i = x_i P$. The vehicle encrypts its real identity and address information using the public key U_j and then sends the ciphertext $EA_i = Enc(U_j, addr_i, id_i)$ to the CS_j . Upon receipt of a registration request from vehicle V_i , CS_j initially assesses the vehicle's reputation score Re_i . Should the reputation value surpass the system's predefined threshold, indicating that V_i is considered legitimate and trustworthy, the CS_j sends the signed transaction $TX_j^1 = \{addr_i\}_{a_j}$ to the blockchain. This triggers the smart contract defined by Algorithm 2, enabling V_i to complete the address registration. Next, the vehicle sends the signed transaction of the public key $TX_i^1 = \{X_i\}_{e_i}$ to the blockchain network. This action initiates the smart contract outlined in Algorithm 3, facilitating the registration of the vehicle's public key information securely on the blockchain. In this scheme, public key registration uses a cuckoo filter to store public key fingerprints, significantly reducing storage space compared to storing the full public keys.

D. Request Phase

In the request phase, vehicle V first initiates an AKA request towards the RSU. Upon receiving and verifying the sender's identity, the RSU subsequently forwards this request to the blockchain network. The detailed sequence of events is outlined below:

1) **Vehicle Side:** To initiate the key agreement process, vehicle V_i first selects a random number q_i and computes the corresponding public key $Q_i = q_i P$ for the subsequent key agreement. The vehicle obtains the current timestamp t_i and computes the signature ϵ_i as shown in Equation (13). V_i then sends a message $M_i = \{addr_i, t_i, Q_i, X_i, \epsilon_i\}$ containing the signature information to RSU_k . Here, $addr_i$ and X_i are values obtained during the registration phase.

$$\epsilon_i = q_i + x_i \cdot h(addr_i, Q_i, t_i, X_i) \quad (13)$$

2) *RSU Side*: Upon receiving the message M_i , RSU_k first validates the timestamp and the vehicle reputation value Re_i . If $|t_i - t_{cur}| \leq \Delta T$ or the Re_i value of the vehicle is below a system threshold, RSU_k rejects the message and terminates the process. Where t_{cur} represents the current timestamp and ΔT represents the maximum reasonable time interval. If the message M_i passes the verification of RSU_k , RSU_k selects a random number q_k to compute the corresponding public key $Q_k = q_k P$ to be used in the key agreement. RSU_k then obtains the current timestamp t_k and sends the ECDSA signature transaction $TX_k = \{addr_i, t_i, Q_i, t_k, Q_k, X_i, \epsilon_i\}_{s_k}$ to the consortium blockchain.

E. Response Phase

In this phase, the AKA request transaction sent by the vehicle is processed by the blockchain nodes, and after the signature transaction is authenticated by a certain number of blockchain nodes, the vehicle and the RSU can securely agreement the session key for subsequent communication.

Algorithm 4 Public Key Verification Function

Input: pk_i

Output: *bool*

- 1: $\triangleright$ check whether pk_i exists in the array
 - 2: **if** $vf.lookup(pk_i) == true$ **then**
 - 3: **return true**
 - 4: **else**
 - 5: **return false**
 - 6: **end if**
-

1) *Message Authentication*: Upon receiving the signature transaction TX_k , the server first checks the timestamp of the message and the reputation value Re_i of the vehicle. If $|t_i - t_{cur}| \leq \Delta T$ or $|t_k - t_{cur}| \leq \Delta T$ or the reputation value of the vehicle is lower than the system threshold, the server terminates the agreement process. Secondly, the server uses a signature algorithm to check the validity of the signing transaction TX_k . Thirdly, the server checks whether the public key X_i is valid or not, by calling the smart contract described in Algorithm 4. If the public key is not registered on the blockchain, the server rejects the message, terminating the AKA request. Finally, the server uses Equation (14) to check the validity of the signature ϵ_i , if the equation does not hold, indicating that the request message has been tampered with and not authenticated, the server rejects the message, and the AKA request is aborted. Otherwise, the AKA message authentication process passes.

$$\epsilon_i \cdot P \stackrel{?}{=} Q_i + h(addr_i, Q_i, t_i, X_i) \cdot X_i \quad (14)$$

2) *Token Generation*: Upon passing the authentication process, the servers can confirm the legitimacy of the vehicle's identity and the integrity of the transmitted request message. Therefore, the servers will use their cooperative private keys to generate signature shares for the vehicle in the form of Equation (15), where r_j is random number and $r_j \in Z_q$ and $R_j = r_j P$, and x_k denotes the index k of ES or CS (i.e., $x_k = k$ based on the index configure process in the setup phase). The

Algorithm 5 Token Aggregation Function

Input: $h(TX_k), \{(\sigma_j, R_j, tag)\}$

Output: *bool*

- 1: $\triangleright$ check whether the sender is an ES or CS
 - 2: **if** $msg.sender \text{ not in } esa \text{ or } csa$ **then**
 - 3: **return false**
 - 4: **else**
 - 5: $\triangleright$ *count* is used to store the token count.
 - 6: $\triangleright$ *token* is used to store tokens.
 - 7: **for each** $(\sigma_j, R_j, tag) \in \{(\sigma_j, R_j, tag)\}$ **do**
 - 8: **if** $tag = 0$ **then**
 - 9: $ctr0 = count0[h(TX_k)]$
 - 10: $tokens0[h(TX_k)][ctr0] \leftarrow (\sigma_j^0, R_j^0, tag)$
 - 11: $count0[h(TX_k)] \leftarrow ctr0 + 1$
 - 12: **else**
 - 13: $ctr1 = count1[h(TX_k)]$
 - 14: $tokens1[h(TX_k)][ctr1] \leftarrow (\sigma_j^1, R_j^1, tag)$
 - 15: $count1[h(TX_k)] \leftarrow ctr1 + 1$
 - 16: **end if**
 - 17: **end for**
 - 18: **return true**
 - 19: **end if**
-

CS_j generates two signatures σ_j^0, σ_j^1 , for the vehicle using the two cooperative private keys $sk_{0,j}$ and $sk_{1,j}$, while the ES_l generates a signature share σ_l^1 for the vehicle using the cooperative private key $sk_{1,l}$. Each signature share corresponds to the tag value tag , to differentiate signatures associated with different collaborative private keys. Finally, server sends a signature transaction $TX_2^j = \{h(TX_k), \{(\sigma_j, R_j, tag)\}_{s_j}\}$ to the blockchain, triggering the smart contract defined by Algorithm 5.

$$\sigma_j = r_j + \left(sk_j \prod_{k \neq j}^{1 \leq k \leq t} \frac{0 - x_k}{x_j - x_k} \right) \cdot h(addr_i, addr_k, PK, h(TX_k)) \quad (15)$$

Algorithm 6 Token Validation Function

Input: $h(TX_k)$

Output: $(token[])$

- 1: $ctr0 = count0[h(TX_k)]$
 - 2: $ctr1 = count1[h(TX_k)]$
 - 3: **if** $ctr1 < t_2$ or $ctr0 < t_1$ **then**
 - 4: **return NULL**
 - 5: **else**
 - 6: $\triangleright$ return all of tokens related $h(TX_k)$.
 - 7: **return** $tokens0[h(TX_k)], tokens1[h(TX_k)]$
 - 8: **end if**
-

3) *Key Agreement*: The authentication of the AKA request by the blockchain nodes is considered successful only when the signature transaction receives no fewer than t_2 signature shares from the blockchain nodes, and at least t_1 of those signatures come from the CS. Upon meeting these conditions, V_i and RSU_k can invoke the smart contract described in Algorithm 6. If the smart contract yields a null return value, it signifies that the signature share has not reached the preset

system value. Conversely, a non-null return value indicates that the AKA request has been authenticated by a sufficient number of blockchain nodes. Therefore, V_i and RSU_k can use Equation (16) to verify the legitimacy of TX_k according to the signature information obtained on the blockchain, and if the obtained signature information can satisfy the equation, it means that the legitimacy of TX_k has been sufficiently authenticated in the process of the AKA request. As a result, V_i and RSU_k can agree on session keys for subsequent communication using Equation (17).

$$\begin{aligned} & \left(\sum_{j=1}^{t_1} \sigma_j^0 + \sum_{j=1}^{t_2} \sigma_j^1 \right) \cdot P \\ &= \left(\sum_{j=1}^{t_1} \left(sk_j \prod_{k \neq j}^{1 \leq k \leq t_1} \frac{0 - x_k}{x_j - x_k} \right) + \sum_{j=1}^{t_2} \left(sp_j \prod_{k \neq j}^{1 \leq k \leq t_2} \frac{0 - x_k}{x_j - x_k} \right) \right) \\ & \quad \cdot h(addr_i, addr_k, PK, h(TX_k)) \cdot P + \sum_{j=1}^{t_1} R_j^0 + \sum_{j=1}^{t_2} R_j^1 \\ &= \sum_{j=1}^{t_1} R_j^0 + \sum_{j=1}^{t_2} R_j^1 + h(addr_i, addr_k, PK, h(TX_k)) \cdot PK \quad (16) \end{aligned}$$

$$sk_{ik} = h(q_i \cdot Q_k, h(TX_k)) = h(q_k \cdot Q_i, h(TX_k)) \quad (17)$$

V. SECURITY ANALYSIS

A. Correctness

Here, we provide the proof for Equation (16). The proof process cleverly employs the principle of Lagrange interpolation polynomials to ensure that the secret value of each server can be reconstructed using the individual signature shares from different cooperative private keys. Based on this, the system public key can be successfully derived and the correctness of the whole scheme is verified accordingly. It's worth noting that we use $h(M)$ instead of $h(addr_i, addr_k, PK, h(TX_k))$ for the sake of simplifying the proof procedure.

$$\begin{aligned} & \left(\sum_{j=1}^{t_1} \sigma_j^0 + \sum_{j=1}^{t_2} \sigma_j^1 \right) \cdot P \\ &= \sum_{j=1}^{t_1} \left(sk_j \prod_{k \neq j}^{1 \leq k \leq t_1} \frac{0 - x_k}{x_j - x_k} \right) \cdot h(M) \cdot P + \sum_{j=1}^{t_1} R_j^0 \\ & \quad + \sum_{j=1}^{t_2} \left(sp_j \prod_{k \neq j}^{1 \leq k \leq t_2} \frac{0 - x_k}{x_j - x_k} \right) \cdot h(M) \cdot P + \sum_{j=1}^{t_2} R_j^1 \\ &= \left\{ \sum_{i=1}^{t_1} h_1(x_i) \prod_{k \neq i}^{1 \leq k \leq t_1} \frac{0 - x_k}{x_i - x_k} + \sum_{i=1}^{t_1} h_2(x_i) \prod_{k \neq i}^{1 \leq k \leq t_1} \frac{0 - x_k}{x_i - x_k} \right. \\ & \quad + \dots + \sum_{i=1}^{t_1} h_{n1}(x_i) \prod_{k \neq i}^{1 \leq k \leq t_1} \frac{0 - x_k}{x_i - x_k} \left. \right\} \cdot h(M) \cdot P + \sum_{j=1}^{t_1} R_j^0 \\ & \quad + \left\{ \sum_{i=1}^{t_2} p_1(x_i) \prod_{k \neq i}^{1 \leq k \leq t_2} \frac{0 - x_k}{x_i - x_k} + \sum_{i=1}^{t_2} p_2(x_i) \prod_{k \neq i}^{1 \leq k \leq t_2} \frac{0 - x_k}{x_i - x_k} \right. \\ & \quad + \dots + \sum_{i=1}^{t_2} p_n(x_i) \prod_{k \neq i}^{1 \leq k \leq t_2} \frac{0 - x_k}{x_i - x_k} \left. \right\} \cdot h(M) \cdot P + \sum_{j=1}^{t_2} R_j^1 \end{aligned}$$

$$\begin{aligned} &= (a_1 + a_2 + \dots + a_{n1}) \cdot h(M) \cdot P + \sum_{j=1}^{t_1} R_j^0 \\ & \quad + (b_1 + b_2 + \dots + b_n) \cdot h(M) \cdot P + \sum_{j=1}^{t_2} R_j^1 \\ &= \sum_{j=1}^{t_1} R_j^0 + \sum_{j=1}^{t_2} R_j^1 + h(addr_i, addr_k, PK, h(TX_k)) \cdot PK \quad (18) \end{aligned}$$

B. Security Model

Here, a security model is defined to show the interactions between challengers and adversaries and the behaviors of the adversary. Four types of participants are involved in our proposed scheme, i.e., the V_i , the RSU_k , the CS_j , and the ES_l . A series of games are defined and adversary $\mathcal{A}$ and challenger C interact with each other under the rules in these games. We use Π_{Λ}^i to describe the instance of these participants, where i denote the index and $\Lambda \in \{V_i, RSU_k, CS_j, ES_l\}$. We described the details as follows.

- 1) *Execute*(Π_V^i, Π_{RSU}^k): In this oracle, a passive attack can be simulate, which mean that the adversary can get the transmitted messages between Π_V^i and Π_{RSU}^k form the challenger C after sending this query.
- 2) *Send*(Π_{Λ}^i, m_t): In this oracle, an active attack can be simulate, which mean that the adversary will receive the corresponding response from the challenger C after sending the request message m_t for the instance Π_{Λ}^i .
- 3) *Reveal*(Π_V^i, Π_{RSU}^k): In this oracle, the forward security can be simulate, which mean that the adversary will get the generated session key between the Π_V^i and Π_{RSU}^k from the challenger C .
- 4) *Corrupt*(Π_{Λ}^i): In this oracle, the forward security can be simulate, which mean that the adversary will get the long-term secret key (the secret key of blockchain address) from the challenger C .
- 5) *Test*(Π_V^i, Π_{RSU}^k): Upon receiving this query, the challenger C will check the freshness of Π_V^i and Π_{RSU}^k . If one of them has been requested, C will abort this query. Otherwise, C will choose a number b in a random toss, where $b \in \{0, 1\}$. If $b = 0$, C will choose a random number as the key to send to $\mathcal{A}$. If when $b = 1$, C will send the key of the session between Π_V^i and Π_{RSU}^k to $\mathcal{A}$.

Definition 1 (Participants): The two instances Π_{Λ}^i and Π_{Λ}^k are said to be participants if they communicate with each other for negotiating a session key in the same session.

Definition 2 (Freshness): If both the instance Π_{Λ}^i and its participant Π_{Λ}^k has not been requested *Reveal*($\cdot$) query, that we can said both Π_{Λ}^i and Π_{Λ}^k are fresh.

Definition 3 (Correctness): The two instances Π_{Λ}^i and Π_{Λ}^k of the same session satisfy correctness if each of them independently generates non-empty and equal session keys.

Definition 4 (Session Key Semantic Security): To complete the game described above, adversary $\mathcal{A}$ need to launch the *Test*($\cdot$) query and submit the guess bit b^* . We assume that the event in which $\mathcal{A}$ happens to win the game i.e. $b^* = b$ is W . The probability that $\mathcal{A}$ wins the game is defined as $Adv(\mathcal{A}) = 2|Pr(W) - 1/2|$ i.e., the definition of semantic safety.

Here, the AKA protocol P is considered semantically secure if the probability $\text{Adv}(\mathcal{A})$ is negligible for any polynomial-time constrained probabilistic.

C. Proven Security

Theorem 1: According to the above security model, the proposed AKA protocol meets the session key semantic security if no adversary with a polynomial time and bounded chance can break ECDDHP.

Proof: In this proof, we design a sequence of games where the adversary $\mathcal{A}$ and the challenger $\mathcal{C}$ interact based on predefined rules. The case that $\mathcal{A}$ wins the i th game is marked as W_i . Additionally, the following are the definitions of these games.

Game 0. In game 0, two tuples $\{A = aP, B = bP, C = abP\}$ and $\{A = aP, B = bP, C = cP\}$, where a, b, c are random numbers in Z_q , are provided to challenger $\mathcal{C}$. The challenger $\mathcal{C}$ will attempt to break ECDDHP with the assistance of $\mathcal{A}$. It is noted that $\mathcal{A}$ can request the oracles according to the following definitions.

- 1) $\text{Send}(\Pi_V^i, \Pi_{RSU}^k, \text{reg})$: Upon receiving this query, $\mathcal{C}$ determines if $KV_1[\Pi_V^i, \Pi_{RSU}^k]$ is null. $\mathcal{C}$ returns $KV_1[\Pi_V^i, \Pi_{RSU}^k]$ to $\mathcal{A}$ if it is not null. If not, $\mathcal{C}$ uses the protocol method to generate $t_i, e_i, \text{addr}_i, X_i$, and TX_i^1 for Π_V^i and t_k, Q_k, TX_k for Π_{RSU}^k . It then stores $\{t_i, e_i, \text{addr}_i, X_i, TX_i^1, t_k, Q_k, TX_k\}$ as the value of $KV_1[\Pi_V^i, \Pi_{RSU}^k]$ and sends the value tuple to $\mathcal{A}$.
- 2) $\text{Send}(\Pi_V^i, \Pi_{RSU}^k, \{X_i\})$: When $\mathcal{C}$ receives this query, it first checks whether tuple $KV_1[\Pi_V^i, \Pi_{RSU}^k]$ is empty. If it is empty, $\mathcal{C}$ terminates this query. Otherwise, $\mathcal{C}$ selects q_i, q_k to compute Q_i, Q_k , uses the value X_i in tuple $KV_1[\Pi_V^i, \Pi_{RSU}^k]$ to compute the signature ϵ_i and then generates the ECDSA signature transaction TX_k , and finally sends TX_k to $\mathcal{A}$.
- 3) $\text{Send}(\Pi_V^i, \Pi_{RSU}^k, \{TX_k\})$: When $\mathcal{C}$ receives this query, it first verifies the validity of the transaction TX_k as well as Equation (14). If there is a validation failure, $\mathcal{C}$ terminates this query. Otherwise, $\mathcal{C}$ generates t_2 tokens $\{(\sigma_j, \text{tag})\}$ generated by the ES s and CS s and stores them in the tuple $KV_2[\Pi_V^i, \Pi_{RSU}^k]$. Finally, these tokens are sent to $\mathcal{A}$.
- 4) $\text{Send}(\Pi_V^i, \Pi_{RSU}^k, \{TX_k, \{(\sigma_j, R_j, \text{tag})\} (1 \leq j \leq t_2)\})$: When $\mathcal{C}$ receives this query, it first checks the validity of these shares using Equation (16). If the validation fails, $\mathcal{C}$ aborts this query. Otherwise, $\mathcal{C}$ computes the session key sk_{ik} and stores it in tuple $KV_3[\Pi_V^i, \Pi_{RSU}^k]$, and the instances Π_V^i and Π_{RSU}^k are set to the unfreshed state.
- 5) $\text{Execute}(\Pi_V^i, \Pi_{RSU}^k)$: When $\mathcal{C}$ receives this query, it first checks the values in tuples $KV_1[\Pi_V^i, \Pi_{RSU}^k]$ and $KV_2[\Pi_V^i, \Pi_{RSU}^k]$. If neither tuple is empty, $\mathcal{C}$ sends the values in these tuples to $\mathcal{A}$; otherwise, $\mathcal{C}$ terminates this query.
- 6) $\text{Corrupt}(\Pi_V^i, \Pi_{RSU}^k)$: When $\mathcal{C}$ receives this query, it first checks the value in tuple $KV_3[\Pi_V^i, \Pi_{RSU}^k]$. If the tuple is not empty, $\mathcal{C}$ sends the value in the tuple to $\mathcal{A}$; otherwise, $\mathcal{C}$ terminates this query.
- 7) $\text{Test}(\Pi_V^i, \Pi_{RSU}^k)$: When $\mathcal{C}$ receives this query, it first checks the freshness of instances Π_V^i and Π_{RSU}^k . If it

passes the check. $\mathcal{C}$ will toss a coin $b \in \{0, 1\}$ and if $b=1$, $\mathcal{C}$ sends the session key sk_{ik} negotiated by instances Π_V^i and Π_{RSU}^k to $\mathcal{A}$. If $b=0$, then $\mathcal{C}$ will select a random number to send to $\mathcal{A}$.

The rules in game 0 correspond to the protocols defined in the security model without any modification. Therefore, the probability that the adversary $\mathcal{A}$ wins the game is satisfied:

$$\text{Adv}_{\mathcal{A}} = 2|\text{Pr}(W_0) - 1/2| \quad (19)$$

Game 1. Game 1 and Game 0 are exactly the same, with the exception that Game1 simulates hash oracles. More specifically, after receiving the request m_i for the hash function $h(\cdot)$, $\mathcal{C}$ checks whether or not $h(m_i)$ is empty. If so, $\mathcal{C}$ selects a random number r_i as the $h(m_i)$ value and sends it to $\mathcal{A}$. If not, $\mathcal{C}$ gives $h(m_i)$ back to $\mathcal{A}$. Clearly, Game 1 and Game 0 are indistinguishable, so the following equation is satisfied.

$$\text{Pr}(W_1) = \text{Pr}(W_0) \quad (20)$$

Game 2. To prove it, we construct the simulation where the number of involved servers equals to t_2 , i.e., the involved servers are $CS_1, CS_2, \dots, CS_{t_1}, ES_{t_1+1}, \dots, ES_{t_2}$. Among these servers, we assume that the adversary $\mathcal{A}$ controls $CS_2, \dots, CS_{t_1}, ES_{t_1+1}, \dots, ES_{t_2}$ whereas the forger $\mathcal{F}$ controls the honest participant CS_1 . Based on the input of public key $PK = K + U$ for the Schnorr signature algorithm, $\mathcal{F}$ interacts with $\mathcal{A}$ using the following simulations.

- 1) **Initialization:** a) $\mathcal{F}$ chooses randoms $u_1 = \{a_1, b_1\}$ where a_1 and $b_1 \in Z_q^*$, calculates $PK_1 = \{U_1 = a_1P, K_1 = b_1P\}$ and $f_1(x) = \{h_1(x), p_1(x)\}$, and sends public material $\{(PK_1, f_1(j)) | 2 \leq j \leq t_2\}$ to $\mathcal{A}$. b) $\mathcal{A}$ is required to send $\{(f_j(1), PK_j) | 2 \leq j \leq t_2\}$ to $\mathcal{F}$. c) Based on the public key $PK = K + U$, $\mathcal{F}$ rewinds $\mathcal{A}$ to step a) for replacing U_1 and K_1 with $\hat{U}_1 = U - \sum_{j=2}^{t_1} U_j$, $\hat{K}_1 = K - \sum_{j=2}^{t_1} K_j$, so that the public material received by $\mathcal{A}$ is $\{(\hat{PK}_1, f_1(j)) | 2 \leq j \leq t_2\}$.
- 2) **Token Generation:** a) For the token generation query for the AKA request message m , $\mathcal{F}$ chooses random elements $R_1 = \{R_1^0, R_1^1\}$ where $R_1^0, R_1^1 \in \mathbb{G}$ and $\sigma_1 = \{\sigma_1^0, \sigma_1^1\}$ where σ_1^0 and $\sigma_1^1 \in Z_q^*$ and sends them to $\mathcal{A}$. b) $\mathcal{A}$ is required to send $(R_j, \sigma_j) | 2 \leq j \leq t_2$ to $\mathcal{C}$. c) $\mathcal{C}$ sends m to the Schnorr signature oracle to get (σ, R) , calculates $\hat{R}_1^0 = R_A - \sum_{j=2}^{t_1} R_j^0$, $\hat{R}_1^1 = R_B - \sum_{j=2}^{t_1} R_j^1$, $\hat{\sigma}_1^0 = \sigma_A - \sum_{j=2}^{t_1} \sigma_j^0$ and $\hat{\sigma}_1^1 = \sigma_B - \sum_{j=2}^{t_1} \sigma_j^1$ and rewinds $\mathcal{A}$ to step a) for replacing (R_1, σ_1) with $(\hat{R}_1, \hat{\sigma}_1)$. Here R_A denotes $\sum_{j=1}^{t_1} R_j^0$, R_B denotes $\sum_{j=1}^{t_1} R_j^1$, σ_A denotes $\sum_{j=1}^{t_1} \sigma_j^0$, and σ_B denotes $\sum_{j=1}^{t_1} \sigma_j^1$.

We can observe that the aforementioned simulation does not affect the validity of the generated tokens, as Equation (21) holds, satisfying the verification described in Equation (16). The only distinction between the real and simulated views is the substitution of U_1, K_1, R_1^0 and R_1^1 with $\hat{U}_1, \hat{K}_1, \hat{R}_1^0$ and $\hat{R}_1^1$, respectively, which is computationally indistinguishable from the perspective of $\mathcal{A}$. Therefore, we can construct a distinguisher such that the probability of distinguishing Game 2 and Game 1 is equal to the probability of breaking Schnorr and ECDSA signature algorithms. As proven in [24], [25], the advantages ϵ_1 and ϵ_2 of breaking ECDSA and Schnorr

signature algorithms by a PPT-adversary under a random oracle satisfy Equations (22) and (23), where q_s denotes the number of requests to the $Send(\cdot)$ oracle, and ϵ represents the probability of breaking ECDLP, thus deducing Equation (24).

$$\begin{aligned} & \left(\sum_{j=1}^{t_1} \sigma_j^0 + \sum_{j=1}^{t_2} \sigma_j^1 \right) \cdot P \\ &= \left(\sigma_A - \sum_{j=2}^{t_1} \sigma_j^0 + \sum_{j=2}^{t_1} \sigma_j^0 + \sigma_B - \sum_{j=2}^{t_2} \sigma_j^1 + \sum_{j=2}^{t_2} \sigma_j^1 \right) \cdot P \\ &= (\sigma_A + \sigma_B) \cdot P \\ &= \sum_{j=1}^{t_1} R_j^0 + \sum_{j=1}^{t_2} R_j^1 + h(PK, m) \cdot (u + k)P \end{aligned} \quad (21)$$

$$\epsilon_1 \leq \frac{3q_s(q_s + q_h)}{(q-1)/2 - q_s - q_h} + (q_s + q_h)^2/2^q + q_h\epsilon \quad (22)$$

$$\epsilon_2 \leq \sqrt{\frac{q + 2q_s(q_s + q_h) + q(q_s + q_h)\epsilon}{q}} \quad (23)$$

$$|Pr(W_2) - Pr(W_1)| \leq \epsilon_1 + \epsilon_2 \quad (24)$$

Game 3. Game 3 is identical to Game 2 except for a slight modification in the $Send(\cdot)$ oracle, where $\mathcal{C}$ selects a random index $idx \in q_s$ and substitutes Q_i , Q_k , and Q_{ik} with A , B , and C . Assuming we can construct a distinguisher capable of discerning between Game 3 and Game 2, it suggests that $\mathcal{C}$ can employ $\mathcal{A}$ as a subroutine to breach ECDDHP. Hence, the Equation (25) holds, where ϵ^* represents the probability of breaking ECDDHP.

$$|Pr(W_3) - Pr(W_2)| \leq q_s \cdot \epsilon^* \quad (25)$$

In Game 3, $\mathcal{A}$ cannot effectively differentiate between a randomly generated key and sk_{ik} created using Equation(17) with a significant probability, unless $\mathcal{A}$ successfully guesses $h(q_i \cdot Q_k, h(TX_k))$. The probability of such an event occurring can reach up to q_h/q . Hence, the Equation (26) holds.

$$|Pr(W_3)| \leq \frac{1}{2} + q_h/q \quad (26)$$

Based on above equations, we can derive the Equation (27).

$$\begin{aligned} Adv_{\mathcal{A}} &= 2|Pr(W_0) - 1/2| = 2|Pr(W_1) - 1/2| \\ &\leq 2(|Pr[W_2] - Pr[W_1]| + |Pr[W_3] - Pr[W_2]| \\ &\quad + |Pr[W_3] - \frac{1}{2}|) \\ &\leq 2(\epsilon_1 + \epsilon_2 + q_s \cdot \epsilon^* + q_h/q) \\ &\leq \frac{6q_s(q_s + q_h)}{(q-1)/2 - q_s - q_h} + 2(q_s + q_h)^2/2^q + 2q_h\epsilon \\ &\quad + 2\sqrt{\frac{q + 2q_s(q_s + q_h) + q(q_s + q_h)\epsilon}{q}} \\ &\quad + 2q_s \cdot \epsilon^* + 2q_h/q \end{aligned} \quad (27)$$

D. Security and Privacy Comparisons

In this section, we undertake a comparative evaluation of our proposed scheme and several related schemes, including Vangala et al. scheme [26], Shen et al. scheme [27], Han et al.

scheme [28], and Ma et al. scheme [29] in terms of the satisfied security and privacy requirements. We define R1 (mutual authentication), R2 (confidentiality), R3 (resistance to replay attacks), R4 (resistance to man-in-the-middle attacks), R5 (forward and backward secrecy), R6 (resistance to DDoS attacks), R7 (conditional privacy-preserving), R8 (resistance to collusion attacks), R9 (perfect forward security), R10 (resistance to key exposure attacks), and R11 (required trust level of servers) and Table I presents a comprehensive comparison of security and privacy aspects. Our proposed scheme exhibits higher satisfaction for multiple security and privacy requirements. A detailed comparison of these security and privacy requirements is shown below.

- 1) **R1 (mutual authentication):** Mutual authentication means that communicating parties can mutually verify the integrity and identity validity of their messages before establishing a connection. In the response phase, the receiving entity authenticates the message sender and each message sent by using signature verification algorithm, public key verification algorithm, and reputation check. In the proposed scheme and the comparison schemes [26], [27], [28], [29], the signature verification algorithm ensures that none of the contents of the sent message has been modified. The public key verification algorithm and the reputation check ensure the legal identity of the vehicle and the RSU, and the authentication share sent by the server ensures that the vehicle and the RSU can achieve mutual authentication.
- 2) **R2 (confidentiality):** Confidentiality means that only entities participating in the same AKA session have the ability to correctly compute the session key. In the proposed scheme and the comparison scheme, the entities use their respective private variables to compute the session key dedicated to both parties respectively. Since the adversary does not have access to the user's private variables, it is unable to compute the session key, and only the entities involved in the key agreement have the ability to compute that key, so all of the schemes [26], [27], [28], [29] satisfy confidentiality.
- 3) **R3 (resistance to replay attacks):** The replay attack is when an adversary is unable to intercept the user's legitimate communication data and repeatedly sends these data to launch an attack. In the proposed scheme and the comparison schemes [26], [27], [28], [29], each message M sent by the communicating parties is accompanied by a timestamp T_i . The receiver can check the timeliness of M by checking whether the equation $|T_j - T_{cur}| \leq \Delta T$ holds, where T_{cur} denotes the current timestamp, and ΔT denotes the maximum legitimate time interval. Thus, replay attacks initiated by an attacker repeatedly transmitting outdated information can be prevented in all of the schemes.
- 4) **R4 (resistance to man-in-the-middle attacks):** Man-in-the-middle attack is an attack launched by an active attacker by forging a valid message after intercepting a valid message from a valid sender. The proposed scheme and the comparison schemes [26], [27], [28], [29] use digital signature and pseudonym techniques respectively,

TABLE I
SECURITY AND PRIVACY COMPARISONS

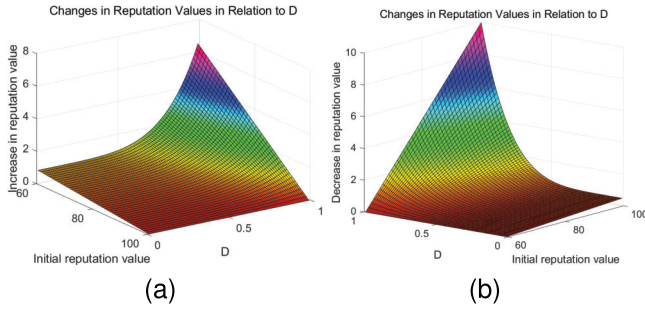
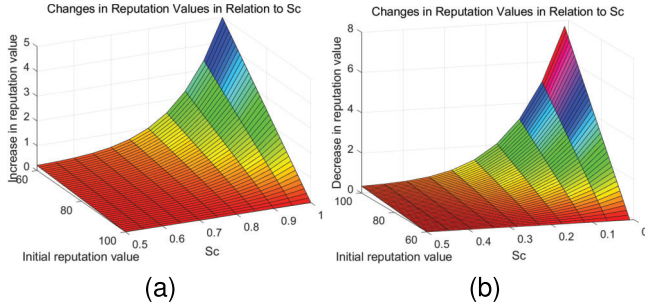
	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11
Vangala et al.'s scheme [26]	✓	✓	✓	✓	✓	×	✓	×	✓	×	⊗
Shen et al.'s scheme [27]	✓	✓	✓	✓	✓	×	✓	×	✓	×	⊗
Han et al.'s scheme [28]	✓	✓	✓	✓	✓	×	×	×	✓	×	⊗
Ma et al.'s scheme [29]	✓	✓	✓	✓	✓	×	✓	×	✓	✓	⊗
Our proposed scheme	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	⊙

✓: The requirement is satisfied ⊗: Require fully-trusted environment
 ×: The requirement is not satisfied ⊙: Require semi-trusted environment

not only the adversary is unable to forge the user's pseudonym information at the same time any tampering of the data by the adversary will result in the information not being able to pass the message authentication, hence these schemes protect against man-in-the-middle attacks.

- 5) **R5 (forward and backward secrecy):** Backward and forward security means that even if the current session key is compromised, the communication data encrypted with the key in past and future sessions will not be compromised. In the proposed and comparison schemes [26], [27], [28], [29], the session key for each session is generated along with random parameters unique to each session. The parameters used in each session are independent of the parameters used in other sessions. Thus, the uniqueness of each session key ensures isolation and prevents any compromise in the security of keys belonging to other sessions.
- 6) **R6 (resistance to DDoS attacks):** DDoS attack means that the attacker sends a large number of false requests to the attacked server so that the server is unable to provide normal service or resource access for a period of time. The proposed scheme establishes a consortium blockchain distributed authentication network maintained by multiple servers, and the security of the system is not affected by the failure of individual servers. In addition, the use of timestamps allows users to assess the freshness of messages and filter out many malicious messages. As a result, DDoS attacks can be effectively prevented. In the comparison schemes [26], [27], [28], [29], the AKA process generally relies on trusted centers that make the system vulnerable to DDoS attacks.
- 7) **R7 (conditional privacy-preserving):** Conditional privacy protection means that only a specific trusted entity can reveal the true identity based on the received pseudo-identity. In the proposed scheme and the comparison scheme [26], [27], [28], [29], the vehicle interacts using pseudo-identity information. The adversary cannot access the true identity of the vehicle. And in case the vehicle exhibits malicious behavior, the trusted entity can decrypt to reveal the true identity of the vehicle, thus achieving conditional privacy protection. In contrast, in Han et al. scheme [28], the vehicle uses the real *ID* for data transmission, which cannot achieve conditional privacy protection.

- 8) **R8 (resistance to collusion attacks):** Collusion attack means the act of two or more malicious nodes jointly disrupting normal nodes and networks by disguising or cooperating with each other. In the proposed scheme, the adversary may try to collude with the consortium blockchain nodes to pass the authentication. However, the design of the scheme's distributed authentication network framework is such that the adversary needs to collude with a certain number of nodes in order to succeed, yet such collusion is not feasible in the real world. Hence, the scheme is effective against collusion attacks whereas the comparison schemes [26], [27], [28], [29] are not able to defend against collusion attacks.
- 9) **R9 (perfect forward security):** Perfect forward secrecy means that the leakage of a long-term session key does not affect the old keys established during the key agreement process. In the proposed scheme and the comparison schemes [26], [27], [28], [29], the vehicle only uses its long term session key to generate messages that can be verified by the receiver. Since the long-term session key does not participate in the key agreement process, perfect forward secrecy can be maintained, ensuring its independence and protection from future communications.
- 10) **R10 (resistance to key exposure attacks):** Key exposure attack means if the key held by the trust infrastructure is compromised or leaked, the security of the system is compromised. The proposed scheme and Ma et al. scheme [29], the private keys of the trust infrastructure do not play a direct role in the key agreement process. Even if these private keys are compromised, the session keys of the protocol are still secure because the compromise does not lead to the disclosure of any information related to the session keys. However, the trust infrastructure private keys in the other comparison scenario are involved in the session key agreement process, so the leakage of the private keys can affect the security of the session keys.
- 11) **R11 (required trust level of servers):** The trust level required by the server means the level of trustworthiness required for the authenticated entities. In the proposed scheme, multiple consortium blockchain entities perform distributed authentication of AKA requests and system security is not compromised in case of individual entities being offline, hence only semi-trusted authentication

Fig. 3. The updating of reputation under different D s.Fig. 4. The updating of reputation under different Sc s.

entities are required. However, the comparison schemes [26], [27], [28], [29] typically use a single authenticated entity to authenticate the AKA request and hence require full trustworthiness for the authenticated entity.

VI. PERFORMANCE ANALYSIS

This section assesses the performance of our proposed scheme by employing various tools to establish a simulation environment. The testbed comprises a PC, equipped with an Intel i7-12700K CPU and 16GB of RAM. We leverage the Miracl cryptography library [30], for accurate measurement of cryptographic operation execution times, and Ganache [31] along with the Truffle framework, a development environment and testing framework for Ethereum, to evaluate blockchain-related performance.

A. Reputation Evaluation

Here, we analyse the reputation value of the scheme and delve into the relationship between the vehicle reputation value and its initial value, the vehicle density D of the environment in which it is located, and the vehicle information trustworthiness score Sc . Fig. 3 shows the amount of increase and decrease of the reputation value for different vehicle densities D when the information credibility score is certain, and Fig. 4 shows the amount of increase and decrease of the reputation value for different reputation scores Sc when the current vehicle density D is certain, respectively.

As can be seen from the above analysis, the reputation value mechanism we designed exhibits two important properties: first, it can effectively limit the rapid growth of the reputation value of entities with higher current reputation values, thus

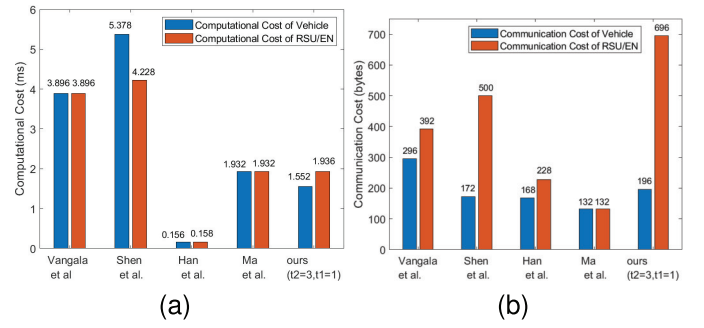


Fig. 5. The comparison of computational/communication costs.

avoiding the rapid accumulation and possible abuse of reputation values. Second, entities with higher reputation values will be penalized more when they publish false information. This design helps prevent entities from abusing their status and publishing inaccurate or misleading information after they have acquired high reputation values. It ensures the stability and effectiveness of the reputation value system and helps to build a fair and trustworthy environment for the vehicular network.

B. Computational Costs Analysis

Fig. 5a appears the computational costs comparison between our proposed scheme and several other AKA schemes proposed in Vangala et al. scheme [26], Shen et al. scheme [27], Han et al. scheme [28], and Ma et al. scheme [29]. We defined T_h , T_{sm-ecc} , T_{eca} , T_{exp} as the execution time of hash function operation, scalar multiplication operation related to elliptic curve, point addition operation related to elliptic curve, modular exponentiation operation, where $T_h \approx 0.002ms$, $T_{sm-ecc} \approx 0.384ms$, $T_{eca} \approx 0.002ms$, $T_{exp} \approx 0.037ms$.

In Vangala et al. scheme [26], the AKA-related computational cost of the vehicle or fog server is $24T_h + 10T_{sm-ecc} + 4T_{eca} \approx 3.896ms$.

In Shen et al. scheme [27], the AKA-related computational cost of the vehicle is $1T_h + 14T_{sm-ecc} \approx 5.378ms$, and the computational cost of the edge node is $2T_h + 11T_{sm-ecc} \approx 4.228ms$.

In Han et al. scheme [28], the computational cost of the vehicle is $4T_h + 4T_{exp} \approx 0.156ms$. Similarly, the computational cost of the server is $5T_h + 4T_{exp} \approx 0.158ms$.

In Ma et al. scheme [29], the computational cost for the vehicle or fog server is approximately $5T_{sm-ecc} + 1T_{eca} + 5T_h \approx 1.932ms$.

In our scheme, the computational cost of the vehicle is $4T_{sm-ecc} + (t_1 + t_2)T_{eca} + 4T_h \approx 1.552ms$. Similarly, the computational cost of the RSU is calculated as $5T_{sm-ecc} + (t_1 + t_2)T_{eca} + 4T_h \approx 1.936ms$.

Based on the above analysis, it shows that our scheme significantly outperforms the Vangala et al. [26], Shen et al. [27], and Ma et al. [29] schemes in terms of computational costs, while it is slightly higher than the scheme of Han et al. [28]. This means that the computational resources occupied by our proposed scheme still have some advantages.

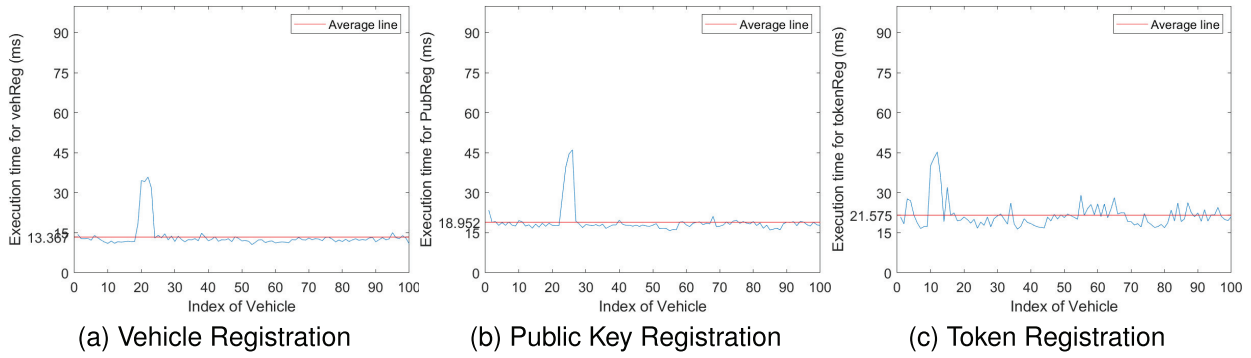


Fig. 6. The processing delays for state-changing Blockchain operations.

C. Communication Costs Analysis

Fig. 5b appears the communication costs comparison between our proposed scheme and several other AKA schemes proposed in Vangala et al. scheme [26], Shen et al. scheme [27], Han et al. scheme [28], and Ma et al. scheme [29]. Our analysis is conducted at a 128-bit security standard, and we choose an additive group G that is generated by a point P with order q over the elliptic curve $E : y^2 = x^3 + ax + b \bmod p$, where p and q are two 256-bit primes and $a, b \in \mathbb{Z}_q$. Importantly, our focus remains restricted to the costs exclusive to the AKA phase, which ensures a more concise and reasonable evaluation.

In the MVFS AKA phase of Vangala et al. scheme [26], in order to establish a session key between the vehicle and the fog node, the communication cost of the vehicle for AKA is $(256 \cdot 7 + 32 \cdot 2 + 512) / 8 = 296$ bytes. The communication cost of the fog node for AKA is $(256 \cdot 10 + 32 \cdot 2 + 512 \cdot 2) / 8 = 392$ bytes.

In the AKA phase of Shen et al. scheme [27], to establish a session key between the vehicle V_i and the edge node, the communication cost on the user side is calculated as $(32 + 512 + 512 + 32 + 32 + 256) / 8 = 172$ bytes. The communication cost on the edge node side is calculated as $(256 + 512 \cdot 3 + 256 \cdot 3 + 32 \cdot 2 + 32 + 256 + 512 + 32 \cdot 2 + 256 \cdot 2) / 8 = 500$ bytes.

In the AKA phase of Han et al. scheme [28], to establish a session key between the vehicle and the server, the communication cost related to the vehicle side is $(32 + 32 + 512 + 512 + 256) / 8 = 168$ bytes. Similarly, the communication cost related to the server side is $(512 + 32 + 512 + 512 + 256) / 8 = 228$ bytes.

In the AKA phase of Ma et al. scheme [29], to establish a session key between the vehicle and the fog server, the communication cost related to the vehicle side is $(512 + 256 + 256 + 32) / 8 = 132$ bytes. The communication cost related to the fog server side is also $(512 + 256 + 256 + 32) / 8 = 132$ bytes.

In the AKA phase of our scheme, a secure session key is established between the vehicle and the RSU, the communication cost for the vehicle in this AKA phase is calculated as $(256 \cdot 2 + 512 \cdot 2 + 32) / 8 = 196$ bytes. The communication cost for the RSU in this AKA phase is calculated as

TABLE II
THE GAS COSTS OF STATE-CHANGING OPERATIONS

Operations	Gas
Contract Deployment	1804391
Filter Initiation	$167010 + 37373 \cdot i + 4812 \cdot b$
Vehicle Registration	46903
Public Key Registration	63237
Token Registration	137730

$(256 \cdot 3 + 32 \cdot 2 + 512 \cdot 3 + (256 + 256 + 256 + 32)t_2 + (256 + 256 + 32)t_1 + 256) / 8 = 328 + 100t_2 + 68t_1$ bytes.

Based on the above analysis, it indicates that our proposed scheme incurs slightly higher communication costs compared to the other schemes, which is a necessary trade-off for achieving full decentralization and thereby significantly enhancing system security and robustness. Nonetheless, our scheme retains notable advantages when evaluated from a holistic perspective.

D. Blockchain-Related Costs Analysis

In terms of evaluating blockchain overhead, we deeply measured the gas cost and processing latency to ensure that our proposed scheme has efficient and practical performance in a VANETs environment. By creating a simulated local blockchain network using Ganache, we wrote smart contracts using Solidity and used the Truffle framework for smart contract deployment and testing.

In the simulation experiment, we modeled a scenario in which 100 cars sent different types of transactions that triggered various operations of the smart contract. Through the Web3 callback function, we accurately captured the receipts of each transaction and recorded the gas consumption and processing latency of each transaction in detail. As illustrated in Table II, we detailed the gas costs of each operation. Meanwhile, in Fig. 6, we visualize the distribution of processing delay. Due to the natural differences in operating system task scheduling and blockchain network transaction processing, the processing delay exhibits some fluctuations among different vehicles. However, through statistical analysis, we derive the average processing latency of vehicle registration,

public key registration, and token registration operations as 13.367 ms, 18.952 ms, and 21.575 ms respectively.

The results of the simulation experiments fully prove that our proposed scheme performs well in terms of processing latency. According to the T/CSAE 53-2020 [32] standard, these latency values are all in line with the maximum latency criterion (500 ms) specified for V2X communication for applications in the high-latency, low-frequency category. Considering that AKA occurs less frequently in real-world applications, the cost of gas for our scheme is found to be reasonable and economical.

VII. CONCLUSION

Here, we propose a hierarchical AKA scheme for VANETs with cloud-edge collaboration powered by consortium blockchain. This innovative scheme leverages consortium blockchain technology, where CSs and ESs function as a robust trust infrastructure, eliminating the need for fully centralized trust centers. Furthermore, we have developed a vehicle reputation system that dynamically adjusts user reputation scores. This mechanism effectively mitigates threats posed by internal attacks, thus safeguarding system security. To ensure resilience against potential security breaches, we also employ a hierarchical threshold voting mechanism. This approach guarantees that the system's security remains intact even in the event of malicious or misbehaving CSs and ESs, significantly enhancing overall security. Our scheme boasts rigorous proof confirming its correctness, security, and adherence to essential security and privacy standards. Furthermore, our performance evaluation underscores its superiority over existing approaches, demonstrating higher computational efficiency, reduced communication overhead, and improved simulation performance.

ACKNOWLEDGMENT

The authors are very grateful to the anonymous referees for their detailed comments and suggestions regarding this article.

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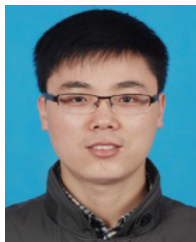


Lu Wei (Member, IEEE) is currently an Associate Professor with the School of Computer Science and Technology, Anhui University. He has over 20 scientific publications in reputable journals (e.g., *IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING*, *IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY*, *IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS*, *IEEE TRANSACTIONS ON MOBILE COMPUTING*, and *IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS*). His research

interests include security and privacy issues in vehicular ad hoc networks, applied cryptography, and blockchain.



Yongjuan Zhang is currently a Research Student with the School of Computer Science and Technology, Anhui University. Her research focuses on the security of the vehicular ad hoc networks.



Jie Cui (Senior Member, IEEE) was born in Henan, China, in 1980. He received the Ph.D. degree from the University of Science and Technology of China in 2012. He is currently a Professor and a Ph.D. Supervisor with the School of Computer Science and Technology, Anhui University. He has over 150 scientific publications in reputable journals (e.g., *IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING*, *IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY*, *IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS*,

IEEE TRANSACTIONS ON MOBILE COMPUTING, *IEEE TRANSACTIONS ON PARALLEL AND DISTRIBUTED SYSTEMS*, *IEEE TRANSACTIONS ON COMPUTERS*, *IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS*, *IEEE TRANSACTIONS ON NETWORK AND SERVICE MANAGEMENT*, *IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS*, *IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS*, *IEEE TRANSACTIONS ON CLOUD COMPUTING*, and *IEEE TRANSACTIONS ON MULTIMEDIA*), academic books, and international conferences. His current research interests include applied cryptography, the IoT security, vehicular ad hoc networks, cloud computing security, and software-defined networking (SDN). He is on the Editorial Board of several international journals, such as *IET Communications*, *Security and Communication Networks*, and *Sensors*.



Hong Zhong was born in Anhui, China, in 1965. She received the Ph.D. degree in computer science from the University of Science and Technology of China in 2005. She is currently a Professor and a Ph.D. Supervisor with the School of Computer Science and Technology, Anhui University. She has over 200 scientific publications in reputable journals (e.g., *IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS*, *IEEE TRANSACTIONS ON PARALLEL AND DISTRIBUTED SYSTEMS*, *IEEE TRANSACTIONS ON MOBILE COMPUTING*, *IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING*, *IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY*, *IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS*, *IEEE TRANSACTIONS ON MULTIMEDIA*, *IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY*, *IEEE TRANSACTIONS ON NETWORK AND SERVICE MANAGEMENT*, *IEEE TRANSACTIONS ON CLOUD COMPUTING*, *IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS*, *IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS*, and *IEEE TRANSACTIONS ON BIG DATA*), academic books, and international conferences. Her research interests include applied cryptography, the IoT security, vehicular ad hoc networks, cloud computing security, and software-defined networking (SDN).



Irina Bolodurina received the Ph.D. degree from South Ural State University. She is currently a Professor and the Head of the Department of Applied Mathematics, Orenburg State University. She has over 60 scientific publications in academic journals and international conferences indexed in Scopus and WoS. She has participated in over 20 scientific projects supported by RFBR and other Russian scientific programs. Her current research interests include theory of optimal control, mathematical modeling, information analysis software, control of

social and economic systems, decision support systems, data integration, and processing.



Debiao He (Member, IEEE) received the Ph.D. degree in applied mathematics from the School of Mathematics and Statistics, Wuhan University, Wuhan, China, in 2009. He is currently a Professor with the School of Cyber Science and Engineering, Wuhan University, and the Shanghai Key Laboratory of Privacy Preserving Computation, MatrixElements Technologies, Shanghai, China. He has published over 100 research papers in refereed international journals and conferences, such as *IEEE TRANSACTIONS ON DEPENDABLE AND SECURE*

COMPUTING, *IEEE TRANSACTIONS ON INFORMATION SECURITY AND FORENSIC*, and the Usenix Security Symposium. His work has been cited more than 10000 times at Google Scholar. His main research interests include cryptography and information security, in particular, cryptographic protocols. He was a recipient of the 2018 IEEE Systems Journal Best Paper Award and the 2019 IET Information Security Best Paper Award. He is on the Editorial Board of several international journals, such as *Journal of Information Security and Applications*, *Frontiers of Computer Science*, and *Human-Centric Computing and Information Sciences*.